

SRS (Software Requirement Specification)

Customer Shopping Behavior Analysis

INTRODUCTION

The **Customer Shopping Behavior Analysis** project analyzes 3,900 customer purchase transactions across various product categories to understand spending behavior, customer preferences, and subscription impact.

The goal of the system is to generate business insights that help improve marketing strategies, customer retention, and revenue growth.

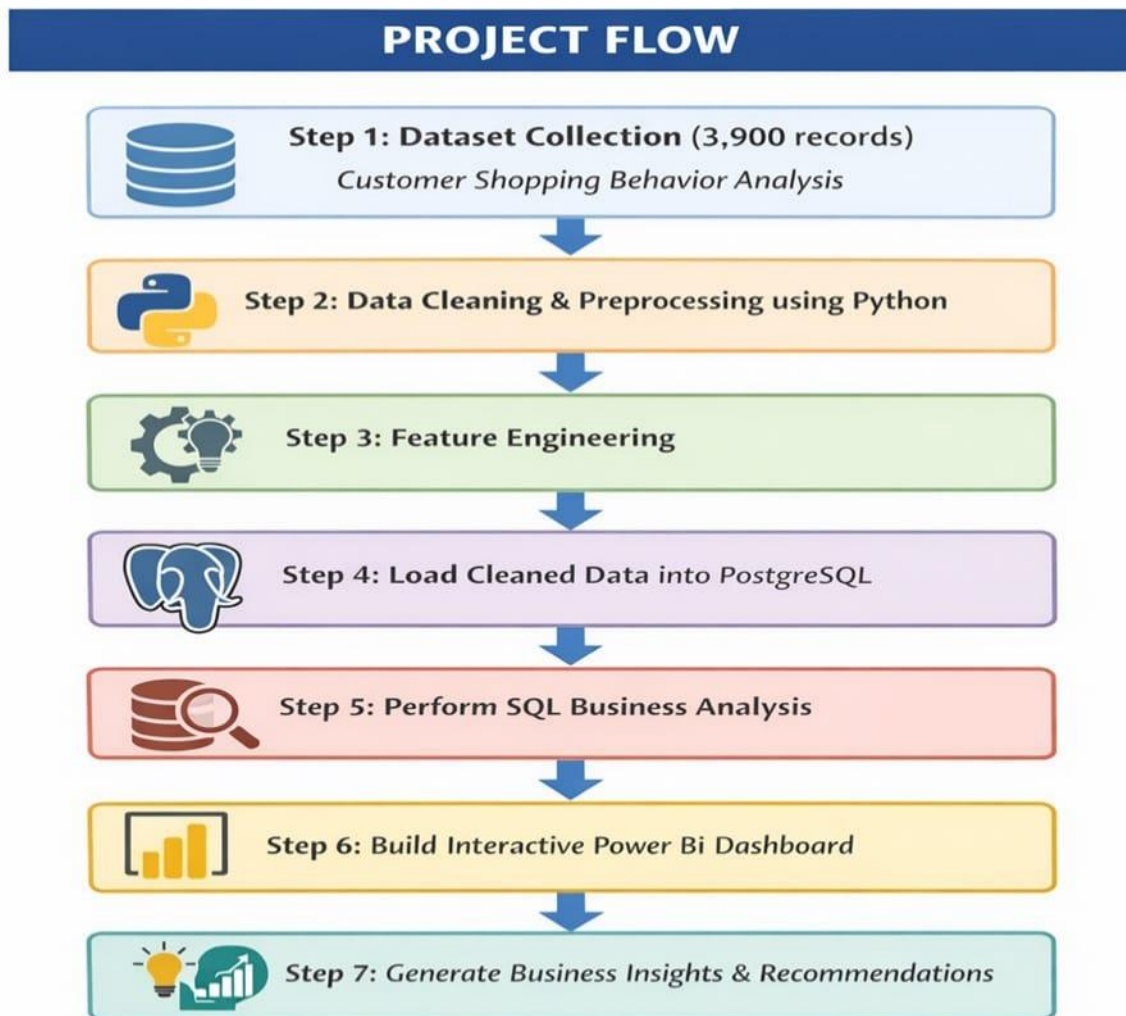
The system uses:

- Python for data cleaning and preprocessing
- PostgreSQL for database storage and SQL analysis
- Power BI for dashboard visualization

PROJECT CONTAINS POINTS

- Data Loading using Python
- Data Cleaning & Missing Value Handling
- Feature Engineering (Age Group, Purchase Frequency)
- Database Integration with PostgreSQL
- SQL Business Queries
- Revenue Analysis (Gender-based revenue)
- Shipping Type Analysis
- Subscription Impact Study
- Customer Segmentation (New, Returning, Loyal)
- Dashboard Development in Power BI
- Business Recommendations

PROJECT FLOW



PROPOSED WORK

The proposed system aims to automate customer shopping behavior analysis by:

- Identifying high-value customers
- Analyzing subscription-based revenue impact
- Studying discount effectiveness
- Segmenting customers into New, Returning, and Loyal categories
- Customer-Shopping-Behavior-Anal...
- Supporting strategic decision-making through dashboards

This system replaces manual analysis with data-driven insights.

MODULES

1.Data Processing Module

- Data Cleaning
- Missing Value Handling
- Feature Engineering

2.Database Module

- Customer Data Storage
- Purchase Transaction Storage
- Subscription Data

3.SQL Analysis Module

- Revenue Analysis
- Product Rating Analysis
- Discount Impact Analysis
- Customer Segmentation

4.Visualization Module

- Revenue Dashboard
- Subscription Analysis
- Shipping Comparison
- Age Group Revenue

5.Reporting Module

- Business Insights
- Strategic Recommendations

BLOCK DIAGRAM

